INCREMENTAL FUZZY CONTROL FOR A BIPED ROBOT BALANCE

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ABSTRACT

A efficient biped robot balance control is essential to achieve dynamic walk. In this work we propose a incremental fuzzy algorithm to control the lateral plane movements for a biped robot. This algorithm is computationally more economic as an adaptive control algorithm. This algorithm demonstrate a smooth balance control response and a correct interaction between the walking algorithm and the lateral plane control. The algorithm was implemented in a PIC16F873 microcontroller and successfully probed in the bipedal robot "Dany walker" designed at the Freie Universität Berlin.

KEY WORDS
Bipedal robots, fuzzy control, artificial intelligence.

1. Introduction

A interesting area of applications for fuzzy controllers is the robotics control, and a relative new application is it use to control biped robots.

Biped robots can walk in almost any type of terrain [1], [2], [3] included those that are impossible for robots with wheels. For this reason, is promising the use of biped robots in human environments as well as the developing of biped robot's control algorithms [4], [5], [6].

Shuuji Kajita [7] designed and developed an almost ideal 2-D model of a biped robot. Kajita supposed for simplicity that the robot's *center of gravity* (*COG*) moves horizontally and he developed a control law for initiation, continuation and termination of the walking process.

Zhen [8] proposed a scheme to enable robot climbing on inclined surfaces. By using force sensors placed in the robot's feet, the transition of the type terrain can be detected, appropriately motors movements are generated to compensate the inclination.

In this paper a fuzzy PD incremental algorithm to control the balance of a biped robot is proposed. This fuzzy control algorithm is computationally economic and allows a condensed implementation. The algorithm was implemented in a PICF873 microcontroller to control the biped robot's balance. This algorithm, was successfully probed achieved dynamic walk in the bipedal robot "Dany walker".

This work is organized as follows. In section 2, the robot's structure is briefly presented. In section 3, the balance control criteria is explained. In section 4, the main lines of the fuzzy PD incremental algorithm are analysed. Finally in section 5, real time results obtained at the bipedal robot are presented.

2. Robot's Structure

The dynamic of a biped robot is closely related with its structure and mass distribution [9], therefore the movement of the *Center of Masses* (*COM*) will have a significant influence on the stability of the robot.

In order to achieve static stability, we place the *COM* as lower as possible. Tu such purpose, a short leg's position was used (figure 1).

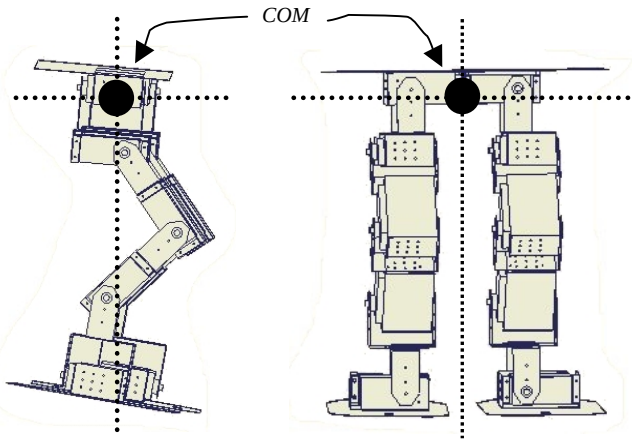


Figure 1. Dany Walker's *COM* location.

To compensate the disturbanceces during walking, lateral movements of the robot were enabled by mechanical design. Thus, was posible to control the lateral balance of the robot by swaying the waist by 4 motors (two at the waist and two at the ankles, figure 2) in lateral plane.

The biped robot used in this work, "Dany Walker" is composed by 10 low-density aluminium links. These are rotational on the pitch axis at the hip, knee and ankle. Each link consists of a modular structure. The links are connected forming a biped robot of 10 degrees of freedom as shown in figure 3.

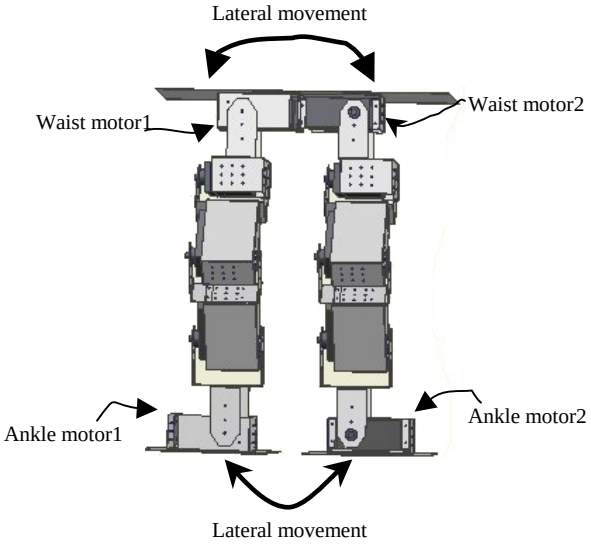


Figure 2. Lateral balance motors.

3. Balance control criteria

Stable walking on biped robots can be characterized by some criteria. Static walking is characterized by maintaining the *COG* inside of the support region. However, the dynamic walking don't need to conserve the *COG* in the support region, the important criteria is to maintain the *Zero Moment Point* (*ZMP*) inside of support region (from now on this criteria will be mentioned as the "*ZMP criteria*"). The use of this criteria has been broadly used to generate biped control algorithms [10], [11].

To achieve stable dynamic walking, the change between simple supports phase and double supports phase should be smooth.

Usually in the beginning of the double supports phase, the foot impact against the floor (when it returns from the air) is very strong and obviously affects the walking balance [12]. In this work we implemented a feedback-force system to calculate the *ZMP*, and feed it in to the fuzzy PD incremental controller to calculate the *ZMP* error. Then the controller adjust the lateral robot's



Figure 3. "Dany Walker" biped robot.

positions to maintain the *ZMP* point inside of the support region.

The robot's balance is achieved by applying the "*ZMP criteria"*. Cubic polynomial interpolation are used to control the sagittal motion. In this paper only the control of the lateral motion (balance) will be discussed.

3.1 Zero moment point (ZMP)

The *ZMP* is a point on the ground where the sum of all momentums are zero. Using this principle, the *ZMP* can be computed as follows:

$$x_{ZMP} = \frac{\sum_i m_i(\ddot{z}+g)x_i - \sum_i m_i \ddot{x} z_i - \sum_i I_{iy}\ddot{\theta}_{iy}}{\sum_i m_i(\ddot{z}+g)} \qquad (1)$$

$$y_{ZMP} = \frac{\sum_i m_i(\ddot{z}+g)y_i - \sum_i m_i \ddot{x} z_i - \sum_i I_{ix}\ddot{\theta}_{ix}}{\sum_i m_i(\ddot{z}+g)} \qquad (2)$$

where $(x_{ZMP}, y_{ZMP}, 0)$ are the *ZMP* coordinates, (x_i, y_i, z_i) is the mass center of the link *i* in the coordinate system, *m* is the mass of the link *i*, and *g* is the gravitational acceleration. I_{ix} and I_{iy} are the inertia moment components, $\ddot{\theta}_{iy}$ and $\ddot{\theta}_{ix}$ are the angular velocity around the axes *x* and *y* (taked as a point from the mass center of the link i).

The force sensors values are direct used to calculate the *ZMP* and for the lateral control is only necessary to know the *ZMP* value for one axis, the *ZMP* calculus is simplified using the formula (3).

$$P_{ZMP} = \frac{\sum_{i=1}^{3} f_i r_i}{\sum_{i=1}^{3} f_i} \qquad (3)$$

where f_i represents the force at the i sensor, and r_i represents the distance between the coordinate origin and the point where the sensor is located. Figure 4, shows the distribution of sensors (marked with tree circles) used for each robot's foot.

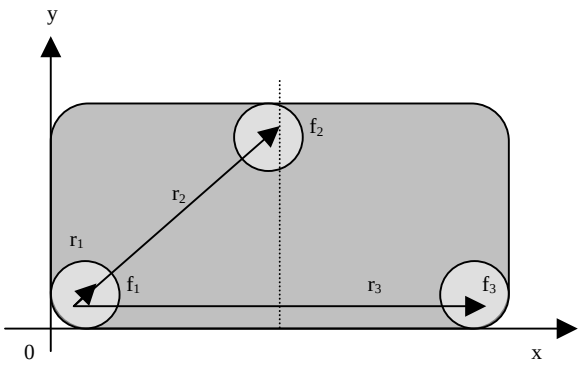


Figure 4. Sensors distribution at the robot's foot.

The total *ZMP* calculus will be obtained by the difference between each foot (4).

$$Total_P_{ZMP} = P_{ZMP1} - P_{ZMP2} \qquad (4)$$

where P_{ZMP1} is the *ZMP* for one foot and P_{ZMP2} is the ZMP for the other.

Figure 5 shows the *ZMP* point (black point) for two robot's standing cases, one before to give a step (left), and other after give a step (right). The pointed line represent the support polygon.

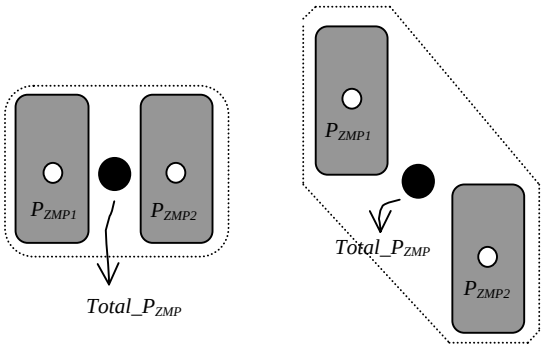


Figure 5. Robots ZMP point (black point) left, before to give a step. Right, and other after give a step.

4. The fuzzy PD incremental algorithm

In this work the fuzzy PD incremental control algorithm is proposed as a variant for the fuzzy PD controller [13] to implement a biped balance control. The fuzzy PD incremental control algorithm consists of only 4 rules and has the structure illustrated in figure 6.

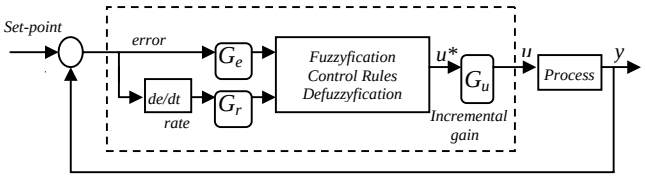


Figure 6. Fuzzy PD incremental algorithm structure.

The gains *Gu*, *Ge* and *Gr* are determined by tuning and they correspond respectively to the output gains, the error (*ZMP error*) and error rate (*ZMP rate*) gains. The value *u** is the defuzzyficated output, or "crisp output".The value *u* is defined by:

$$u = \begin{cases} G_u * (u^*) & if\ |e| < \theta \quad (for \quad t = 0,\ G_{inc} = 0) \\ G_{inc} + inc & if\ |e| > \theta \end{cases} \qquad (5)$$

Where *e* is the error (error*Ge), θ is a error boundary selected by tuning, G_{inc} is the incremental gain obtained adding the increment "*inc*".

Figure 7 shows the flow diagram for the incremental gain of *u* .

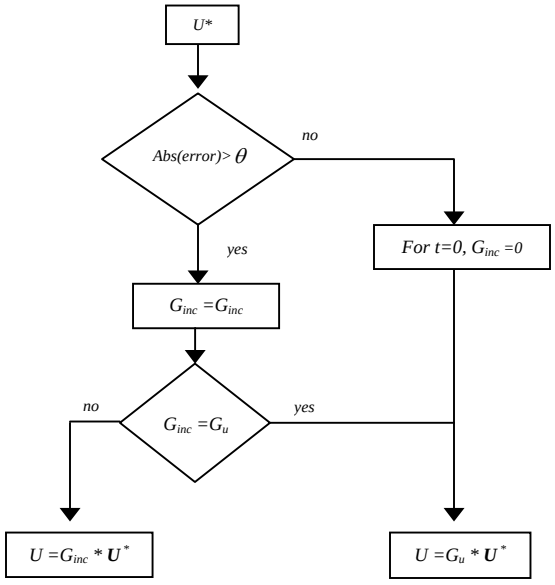


Figure 7. Flow diagram for the incremental gain of Gu.

Figure 8 shows the area where the absolute error is evaluated and the controller output is incremental (u=Gi+inc).

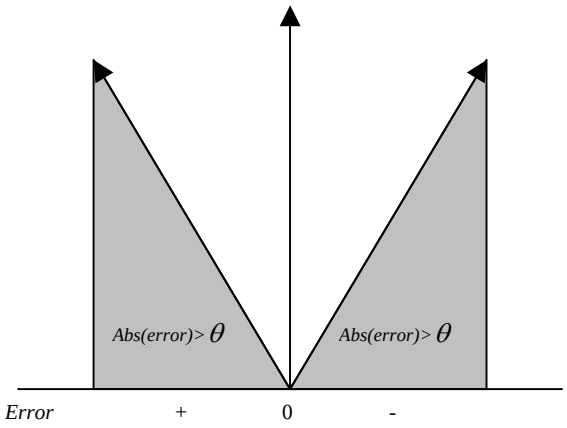


Figure 8. Fuzzy PD incremental absolute error area.

4.1 Fuzzyfication

As is shown in figure 9, there are two inputs to the controller: error and rate. The error is defined as:

$$error = setpoint - y$$

Rate it is defined as it follows:

$$rate = (ce - pe) / sp$$

Where *ce* is the current error, *pe* is the previous error and *sp* is the sampling period. Current and previous error, are referred to an error without gain. The fuzzy controller has a single incremental output, which is used to control the process.

The input an output membership functions for the fuzzy controller are shown in figure 9 and figure 10, respectively. *H* and *L* are two positive constants to be determined. For convenience we will take *H=L* to reduce the number of control parameters to be determined.

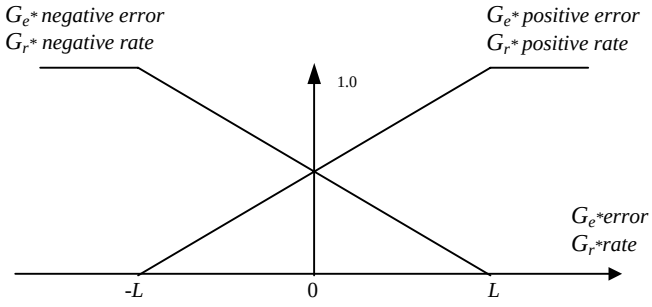


Figure 9. Input membership functions.

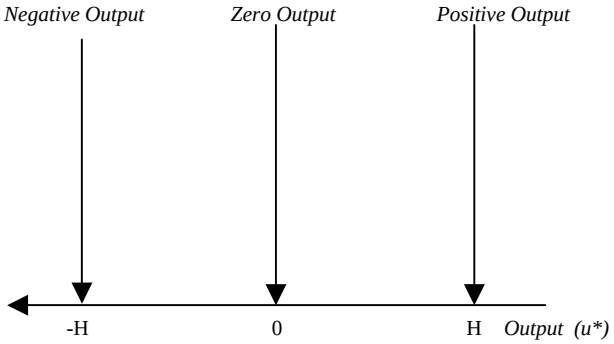


Figure 10. Output membership functions.

The membership functions for the input variables, error and rate, are defined by:

$$\mu_{ep} = \frac{L + (Ge * error)}{2L} \qquad \mu_{en} = \frac{L - (Ge * error)}{2L}$$
$$\mu_{rp} = \frac{L + (Gr * rate)}{2L} \qquad \mu_{rn} = \frac{L - (Gr * rate)}{2L} \qquad (6)$$

4.2 Fuzzy rules and defuzzyfication

Exist four rules to evaluate the fuzzy PD incremental controller [4]:

R1. If *error* is ep and *rate* is rp then output is op

R2. If *error* is ep and *rate* is rn then output is oz

R3. If *error* is en and *rate* is rp then output is oz

R4. If *error* is en and *rate* is rn then output is on

The defuzzyfication method used is the gravity center, in this case is represented by:

$$u = \frac{-H(\mu_{R4(x)}) + 0(\mu_{R2(x)+}\mu_{R3(x)}) + H(\mu_{R1(x)})}{\mu_{R4(x)} + (\mu_{R2(x)+}\mu_{R3(x)}) + \mu_{R1(x)}}$$

5. Real time results

The first test applied to the balance controller was a *x*-direction impulse. The robot was standing in balance and then a disturbance (in the lateral direction) was applied to the biped robot, the figure 11 shows the evolution of the *ZMP* value in centimeters during 10 seconds.

Figure 12 shows the error and rate evolution during the the x-direction impulse application. The incremental PD fuzzy controller's parameters (founded experimentally) were: *Ge*=2, *Gr*=2, and *Gu*=1.

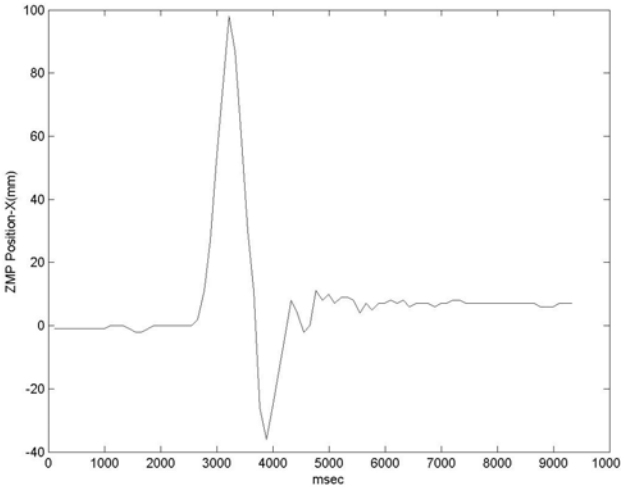


Figure 11. ZMP value evolution achieved by the controller, for a *x*-direction impulse.

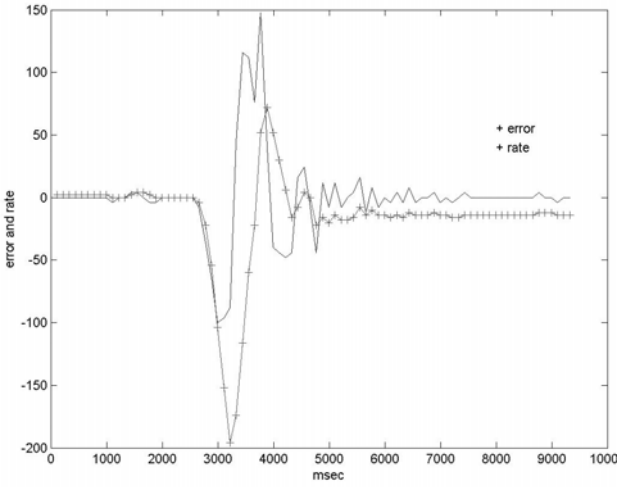


Figure 12. Error and rate evolution for a *x*-direction impulse.

Finally, a walking test balance was applied to the robot. The robot gave some steps in a flat surface, while the balance controller compensated the ZMP value. Figure 13 shows the ZMP value (in centimeters) evolution achieved by the controller during 15 seconds.

Figure 14 shows error and rate evolution during the robots walk. The incremental PD fuzzy controller's parameters (founded experimentally) were: *Ge*=2, *Gr*=2, and *Gu* =1.

The motor's position output was calculed by the controller during the walk. It is showed at Figure 15. This motor's position value is only just for one waist motor.

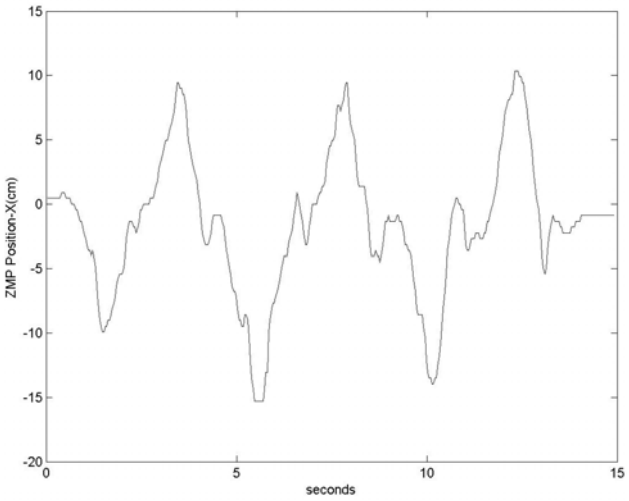


Figure 13. ZMP value evolution achieved by the controller, during the biped walk.

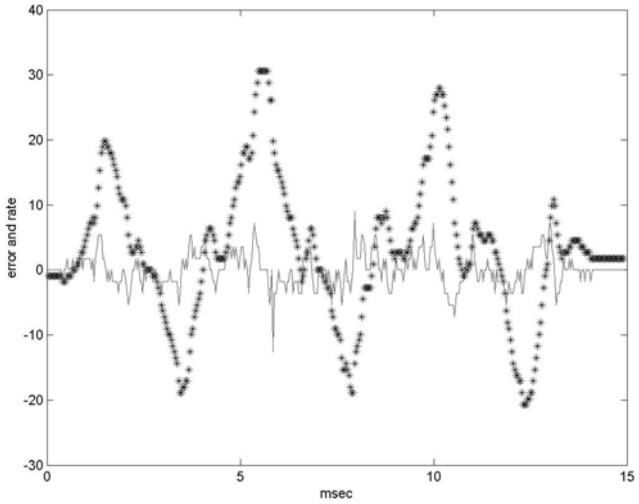


Figure 14. Error and rate evolution during the biped walk.

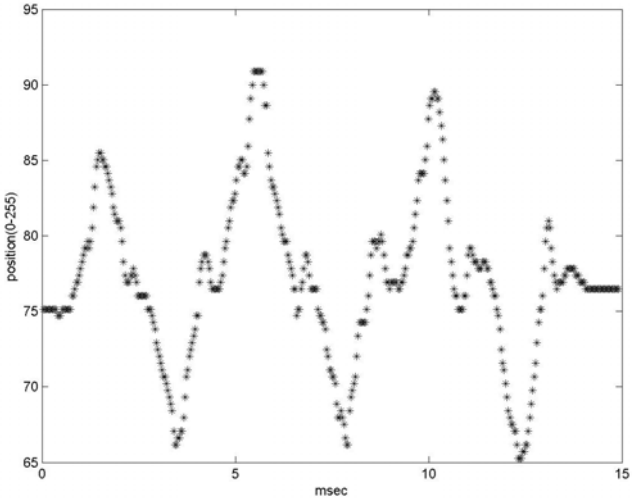


Figure 15. Motor's position output during the biped walk.

Conclusions

From figure 11, is possible to observe the controller's performance for a *x*-direction impulse. The response was fast (approximated 2 seconds), until the controller reach a *ZMP* value near to cero. This feature allows the biped robot to gain stability even during walking (figure 13), maintaining the *ZMP* always inside of the support polygon.

The fuzzy PD incremental controller algorithm demonstrated be appropriated for balance control, and computationally economic (all running in a PIC microcontroller) . The algorithm was successfully probed in real-time at the "Dany walker" biped robot (videos aviable at http://www.inf.fu-berlin.de/~zaldivar).

The algorithm proposed in this paper could be used in other robots structures with a similar dynamic and with other degrees of freedom. For this, will be necessary only adjust the controller's gain parameters. A interesting propose will be the use of a genetic algorithm to find the controller's gain parameters.

A future work will be the implementation of a inclinometer to make data fusion with the *ZMP* value, and obtain more data from the robot's inclination. The goal of the bipedal balance robot control will be to achieve an inclination value of cero and maintain the *ZMP* at the center or at least inside of the support polygon.

The bipedal robot used in this work is part of a biped robot design project which is being developed in the Freie Universität Berlin.

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